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Biochemistry Module, 1st Year Medicine
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METABOLISM OF KETONE BODIES: KETOGENESIS AND KETOLYSIS

course objectives

- Define ketone bodies and where they are produced
- Explain ketogenesis: triggering conditions and the three key reactions
- Describe the benefits of ketogenesis for the liver
- Define ketolysis as the breakdown of ketone bodies into acetyl-CoA.
- Calculate the energy balance of ketolysis
- Explain the regulatory mechanisms of ketolysis
- Distinguish between the physiological benefits of ketogenesis and ketolysis
- Describe the mechanism of diabetic ketoacidosis.

I-INTRODUCTION:

- Ketone bodies serve as **secondary energy substrates** during fasting.
- Once formed, ketone bodies are released into **the bloodstream**.
- They are **soluble** in aqueous solution and **do not need** to be transported as **lipoproteins** or bound to **albumin**, unlike other lipids.
- Produced in the **mitochondrial matrix** of the **liver** from fatty acids, they act as an energy source for peripheral tissues (skeletal muscle, cardiac muscle and renal cortex).
- Use:** to compensate for a lack of glucose in cells (often due to diabetes)
- Metabolism **involves** ketogenesis and ketolysis

II-Ketogenesis

-Ketogenesis is a mechanism **specific to liver** cells and occurs **exclusively** in the **mitochondria**.

-The starting point of ketogenesis is **acetyl-CoA**.

-The liver requires ketogenesis for β -oxidation to function (it consumes a large amount of **coenzyme A** for this synthesis of **acetyl-CoA**).

- The liver **cannot use or store** these ketone bodies; it releases them into the systemic circulation.

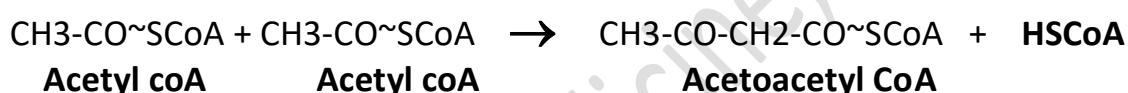
- If the Krebs cycle (KC) is **functioning**, these **CoA** molecules are released and can be reused for **β -oxidation**.
- However, if the KC is **not functioning**, **another mechanism** is required to **regenerate coenzyme A** (CoA).

-**Acetyl-CoA** is converted into **three ketone bodies** (acetoacetate, acetone, and 3-hydroxybutyrate) with regeneration of **CoA**.

There are **three** reactions leading to acetoacetate:

1 – FORMATION OF ACETOACETYL-CoA

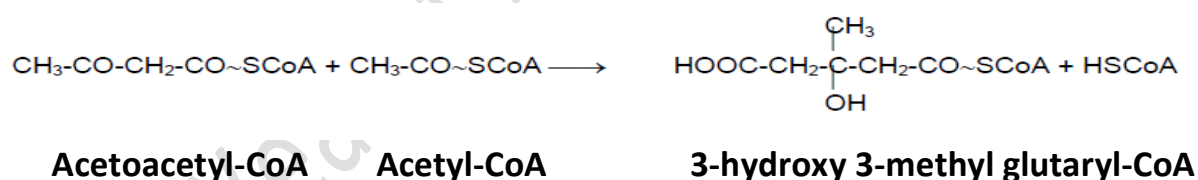
This is catalysed by **acetoacetyl-CoA synthase**



2 – FORMATION OF 3-HYDROXY 3-METHYL GLUTARYL-COA (HMG).

-This compound is also the precursor for **cholesterol synthesis**.

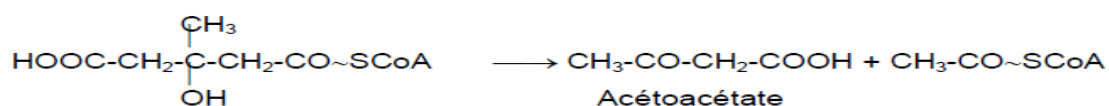
-The reaction is catalysed by **3-hydroxy 3-methyl glutaryl-CoA synthase**, which condenses another acetyl-CoA onto acetoacetyl-CoA.



3 – FORMATION OF KETONE BODIES

3.1 – Formation of acetoacetate

The cleavage of 3-hydroxy-3-methylglutaryl-CoA by **3-hydroxy-3-methylglutaryl-CoA lyase**.



3.2 – Formation of 3-hydroxybutyrate and acetone

Once formed, **acetoacetate** is reduced to **3-hydroxybutyrate** by **3-hydroxybutyrate dehydrogenase** (1) and/or decarboxylated to acetone by **acetoacetate decarboxylase** (2) according to the following reactions:



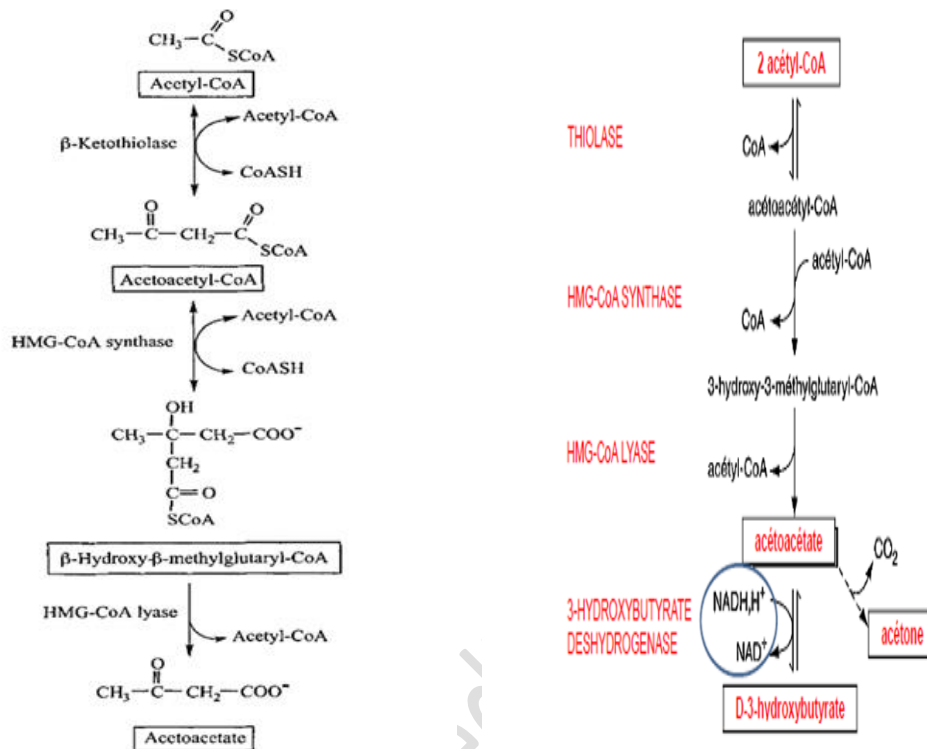
Acetoacetate

3-hydroxybutyrate



Acetoacetate

Acetone



General overview of ketogenesis

III/Ketolysis:

A/ Definition:

-**Ketolysis** is the breakdown of ketone bodies into **acetyl-CoA**, which enters the Krebs cycle to produce ATP.

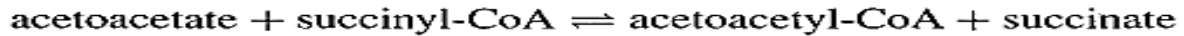
-Ketolysis occurs **in extrahepatic** consumer tissues: primarily skeletal and cardiac muscle, and the brain during prolonged fasting.

-Red blood cells, which lack **mitochondria**, cannot **used** ketone bodies and are therefore **strictly glucose-dependent**.

It also takes place in **four** stages:

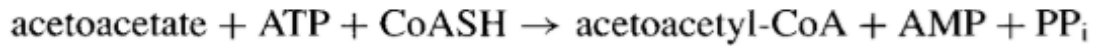
- Reaction 1: Oxidation of β -hydroxybutyrate to acetoacetate, catalyzed by **β -hydroxybutyrate dehydrogenase** (the reverse of reaction 4 in ketogenesis).
- Reaction 2: Activation of acetoacetate to its **coenzyme A ester** via **two** possible mitochondrial reactions:

1.One catalysed by **succinyl CoA transferase (SCOT, a key enzyme)**, which by passes the 5th reaction of the Krebs cycle according to the following reaction scheme:



This reaction results in a net loss of one ATP molecule produced by the 5th reaction of the Krebs cycle.

2. The other is catalyzed by **acetoacetyl-CoA synthetase** according to the following reaction scheme:



This reaction consumes the equivalent of two ATP molecules.

N.B.: Succinyl transferase and acetoacetyl-CoA synthetase are **absent** in **hepatocytes**, hence the liver's inability to oxidise CCs.

- Reaction 3: Thiolysis of acetoacetyl CoA to 2 acetyl CoA, catalysed by a muscle thiolase.
- Reaction 4: The two acetyl-CoA molecules produced by thiolysis enter the Krebs cycle.

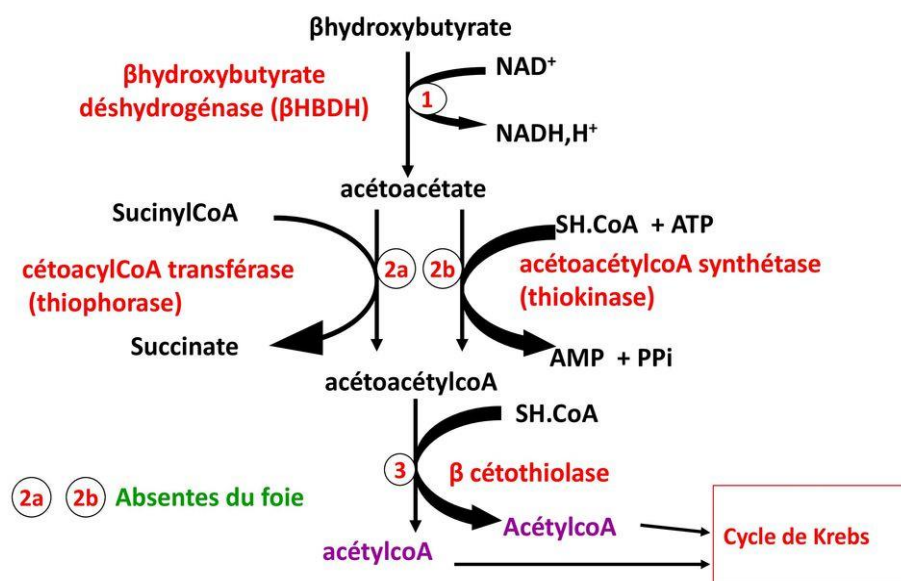
This is because the muscle does **not undergo neoglucogenesis**; its pool of **oxaloacetate** is **preserved** during fasting, allowing the pool of **acetyl-CoA** produced by ketolysis to enter the **Krebs cycle** to produce energy.

-**Acetyl-CoA** is the substrate for **ketogenesis**, which converts it into acetoacetate, β -hydroxybutyrate or acetone.

-These **three** products are secreted into **the plasma** (ketone bodies), then excreted by **the kidneys** (acids) or by **the lungs** (acetone).

-They can also serve as energy substrates for other organs in which the Krebs cycle is active (**brain, heart**).

LES REACTIONS DE LA CETOLYSE



B/Energy Balance of Ketolysis:

Each molecule of β -hydroxybutyrate broken down produces:

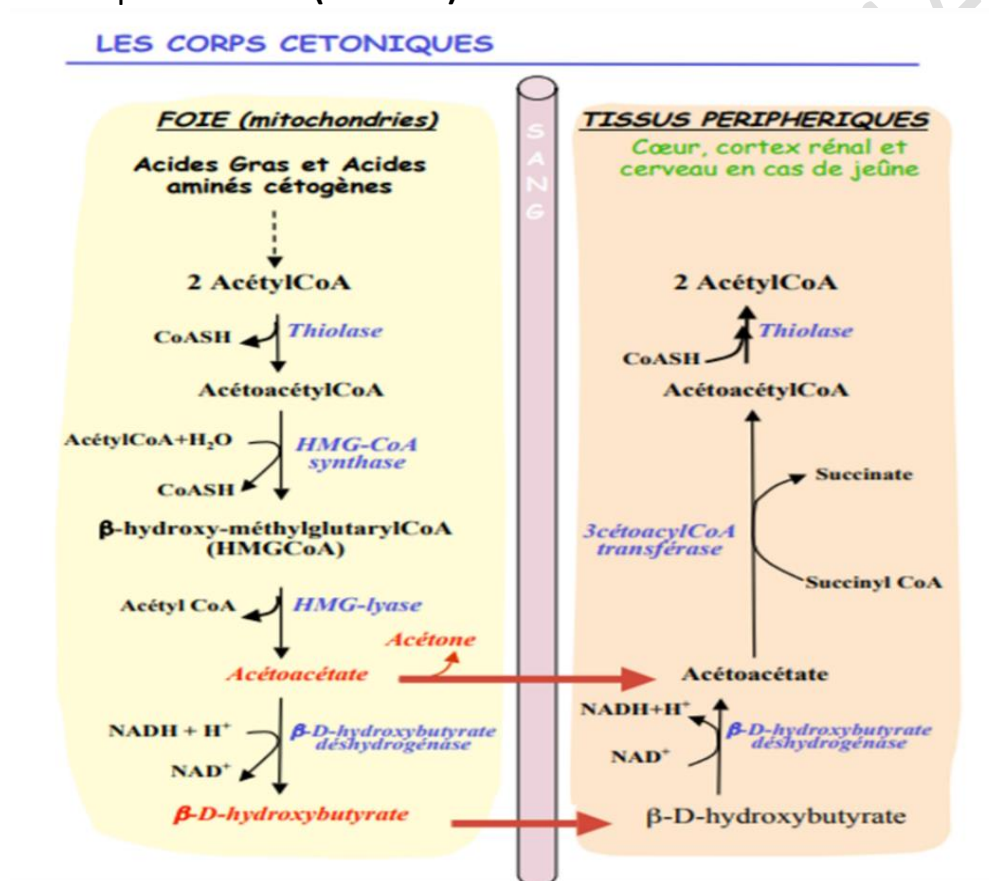
- **2 Acetyl-CoA** → Enter the Krebs cycle (12 ATP per acetyl-CoA) = **24 ATP**.
- **1 NADH** (→ **3 ATP** via the respiratory chain).

Total: 24 + 3 = **27 ATP** per β -hydroxybutyrate.

Compared to **glucose: 38 ATP** per molecule, but ketones are more efficient in the event of glucose shortage.

C/Regulation of Ketolysis:

- Blood ketone levels: Stimulated during fasting, a ketogenic diet or in diabetes.
- Glucose availability: If glucose is abundant, **ketolysis is inhibited** (insulin blocks lipolysis and therefore ketogenesis).
- Enzyme deficiencies: A deficiency in **SCOT** blocks ketolysis → dangerous build-up of ketones (**acidosis**).



IV/The Benefits of Ketogenesis and Ketolysis of Ketone Bodies

1-Main benefits of ketogenesis:

✓ An alternative energy source to glucose:

- Ketone bodies supply the brain, heart and muscles with energy when glucose is in short supply.
- The brain, which normally **depends on glucose**, can use ketones after a few days of fasting (reducing the need for gluconeogenesis).

- Free fatty acids (FFAs) **do not cross the blood-brain barrier** in significant quantities.

- The brain depends mainly on:

- **Glucose** (→ 95% of its energy under normal conditions).

- **Ketone bodies** (in cases of prolonged fasting or a ketogenic diet).

- ✓ **Conservation of muscle protein:**

- Limits muscle breakdown (proteolysis) by providing a non-carbohydrate source of energy.

- ✓ **Metabolic regulation:**

- Reduces inflammation and improves insulin sensitivity (useful in type 2 diabetes).

- ✓ **Optimisation of lipid metabolism:**

- Promotes lipolysis (fat breakdown) and reduces triglyceride storage.

2-Main benefits of ketosis:

- ✓ **Rapid energy during periods of fasting or exercise:**

- Ketones provide up to 70% of the brain's energy after 3 days of fasting.

- More effective than fatty acids for certain tissues (e.g. the brain).

- ✓ **Reduction of oxidative stress:**

- Ketone bodies (especially β -hydroxybutyrate) **have antioxidant** and neuroprotective effects.

- ✓ **Potential therapeutic role:**

- Being studied in neurodegenerative diseases (Alzheimer's, Parkinson's) and certain cancers.

V/Ketogenesis and diabetic ketoacidosis

Diabetes: There is a complete **lack of insulin**.

-Glucose is **abundant** here but **cannot be used** by the body.

-**Lipolysis** therefore occurs, releasing fatty acids (FAs) which enable the **production of acetyl-CoA** in the **liver**, which in turn allows the **formation of ketone bodies**.

-However, the liver **prioritises** producing **glucose** from muscle and amino acids (AAs).

-Excess **ketone bodies** (which are acids) will contribute **to acidosis: diabetic ketoacidosis**.

Conclusion

Ketogenesis and **ketolysis** form a metabolic system essential for **survival** in the **absence of carbohydrates**, with therapeutic benefits (neurology, metabolism) but requiring a balance to avoid complications.